

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415537
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Akishita; Yosuke et al.

Information processing device

Abstract

The information processing device includes a control unit that specifies, for each preset setting area, a setting condition for improving the fuel efficiency of the vehicle by using the low-viscosity oil based on the data of the plurality of vehicles. When the data of the vehicle that does not use the low viscosity oil satisfies the setting condition in the setting area, the control unit executes a notification process for recommending the user of the vehicle that does not use the low viscosity oil to use the low viscosity oil.

Inventors: Akishita; Yosuke (Tokyo, JP), Bandai; Hiroyuki (Nagakute, JP), Sakai; Takurou (Tokyo, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)

Family ID: 1000008818680

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyoto, JP)

Appl. No.: 18/636953

Filed: April 16, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240400080 A1	Dec. 05, 2024

Foreign Application Priority Data

JP	2023-090345	May. 31, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: B60W50/14 (20200101); B60W50/00 (20060101)

U.S. Cl.:

CPC **B60W50/14** (20130101); B60W2050/0083 (20130101); B60W2050/146 (20130101); B60W2530/209 (20200201); B60W2556/45 (20200201)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0225180	12/2006	Tamoto	N/A	N/A
2008/0120175	12/2007	Doering	N/A	N/A
2016/0222867	12/2015	Lee	N/A	B60K 11/085
2018/0273870	12/2017	Burns, III	N/A	C10M 101/02
2023/0174695	12/2022	Schöller	508/469	C10M 145/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2005-344017	12/2004	JP	N/A
2008-129603	12/2007	JP	N/A
2010-188823	12/2009	JP	N/A
6141993	12/2016	JP	N/A

Primary Examiner: Trieu; Van T

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-090345 filed on May 31, 2023, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an information processing device.

2. Description of Related Art

(3) Conventionally, low-viscosity oils that can be used as engine oils have been known (for example, Japanese Unexamined Patent Application Publication No. 2005-344017 (JP 2005-344017 A)).

SUMMARY

(4) When the use of low-viscosity oil improves the fuel efficiency of a vehicle, it is useful to recommend a user of the vehicle to use the low-viscosity oil.

(5) In view of the foregoing, it is an object of the present disclosure to recommend a user of a vehicle whose fuel efficiency is improved by the use of low-viscosity oil to use the low-viscosity oil.

- (6) An aspect of the present disclosure provides an information processing device including a control unit that specifies a setting condition for improving fuel efficiency of a vehicle by using low-viscosity oil based on data on a plurality of vehicles for each setting area set in advance. The control unit is configured to execute a notification process to recommend a user of a vehicle in which the low-viscosity oil is not used to use the low-viscosity oil when data on vehicles in which the low-viscosity oil is not used meet the setting condition in the setting area.
- (7) According to an aspect of the present disclosure, it is possible to recommend a user of a vehicle whose fuel efficiency is improved by the use of low-viscosity oil to use the low-viscosity oil.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
- (2) FIG. 1 is a diagram illustrating a schematic configuration of a system according to an embodiment of the present disclosure;
- (3) FIG. 2 is a sequence diagram illustrating an exemplary operation of the system shown in FIG. 1; and
- (4) FIG. 3 is a diagram illustrating exemplary setting conditions.

DETAILED DESCRIPTION OF EMBODIMENTS

- (5) Hereinafter, an embodiment of the present disclosure will be described below with reference to the drawings.
- (6) As illustrated in FIG. 1, the system 1 according to the present embodiment includes a server device 10 and a plurality of vehicles 20. The server device 10 and the vehicle 20 can communicate with each other via the network 2. The network 2 may be any network including a mobile communication network, the Internet, and the like.
- (7) The server device 10 is an information processing device. The server device 10 is, for example, a dedicated computer configured to function as a server, a general-purpose personal computer, a cloud computing system, or the like.
- (8) The server device 10 is managed by, for example, a vendor of engine oil. The server device 10 has information on the type of engine oil used in each of the plurality of vehicles 20. When low viscosity oil is used for the vehicle 20, the fuel efficiency of the vehicle 20 may be improved. When the fuel efficiency of the vehicle 20 is improved, the vehicle 20 travels a larger distance with a smaller amount of fuel.
- (9) Here, the degree of improvement in the fuel efficiency of the vehicle 20 due to the use of the low-viscosity oil varies from region to region. For example, there are many traffic lights in urban areas compared to suburban areas. Therefore, in the urban area, the frequency of the temporary stop of the vehicle 20 is higher than in the suburban area. When the frequency of the pause of the vehicle 20 increases, the number of times the engine of the vehicle 20 is started increases. The resistance of the engine of the vehicle 20 is particularly high when the engine is started. When the resistance applied to the engine of the vehicle 20 increases, the fuel efficiency of the vehicle 20 deteriorates. When the low viscosity oil is used, the resistance applied to the engine at the time of starting the vehicle 20 can be reduced. That is, in a region including a larger number of urban areas, the degree of improvement in fuel efficiency of the vehicle 20 due to the use of the low-viscosity oil is greater than in the other regions.
- (10) Further, the degree to which the fuel efficiency of the vehicle 20 is improved by the use of the low viscosity oil depends on at least one of the environmental conditions and the driving conditions of the vehicle 20. For example, when the vehicle speed of the vehicle 20 changes, the engine speed

changes. When the rotational speed of the engine changes, the degree to which the fuel efficiency of the vehicle **20** improves due to the use of the low-viscosity oil also changes.

(11) Therefore, the server device **10** specifies, for each of the plurality of setting areas, a setting condition for improving the fuel efficiency of the vehicle **20** by using the low-viscosity oil based on the data of the plurality of vehicles **20**. The set area is set in advance. The setting area may be set by a prefecture or may be set by a local division such as Tohoku or Kanto. Further, as will be described later, the server device **10** specifies the setting condition by specifying the environment information and the traveling information of the vehicle **20** in which the fuel efficiency of the vehicle **20** is improved by the use of the low viscosity oil. Further, in the setting area, when the data of the vehicle **20** that does not use the low-viscosity oil satisfies the setting condition, the server device **10** executes a notification process for recommending the user of the vehicle **20** that does not use the low-viscosity oil to use the low-viscosity oil. By specifying the setting condition for improving the fuel efficiency of the vehicle **20** in each setting area as described above, it is possible to more accurately recommend the use of the low viscosity oil to the user of the vehicle **20** who does not use the low viscosity oil in the setting area.

(12) The server device **10** includes a communication unit **11**, a storage unit **12**, and a control unit **13**.

(13) The communication unit **11** is configured to include at least one communication module that can be connected to the network **2**. The communication module is, for example, a communication module compliant with a standard such as a wired Local Area Network (LAN) or a wireless LAN. The communication unit **11** is connected to the network **2** with the communication module via a wired LAN or a wireless LAN.

(14) The storage unit **12** is configured to include at least one semiconductor memory, at least one magnetic memory, at least one optical memory, or a combination of at least two of them. The storage unit **12** may function as a main storage device, an auxiliary storage device, or a cache memory. The storage unit **12** stores data used for the operation of the server device **10** and data obtained by the operation of the server device **10**. For example, the storage unit **12** stores information on the type of engine oil used in each of the plurality of vehicles **20**.

(15) The control unit **13** is configured to include at least one processor, at least one dedicated circuit, or a combination thereof. The processor may be, for example, a general-purpose processor such as Central Processing Unit (CPU) or Graphics Processing Unit (GPU) or a special-purpose processor specialized for a particular process. The control unit **13** executes processing related to the operation of the server device **10** while controlling each unit of the server device **10**.

(16) The vehicle **20** is, for example, an automobile such as a gasoline-powered vehicle or a hybrid electric vehicle (HEV). However, the vehicle **20** is not limited thereto. Vehicle **20** may be any type of automobile provided with an engine.

(17) The vehicle **20** includes a temperature sensor **21**, an electronic control unit **22**, and an in-vehicle device **30**.

(18) The temperature sensor **21** is capable of detecting the ambient temperature of the vehicle **20**. The ambient temperature of the vehicle **20** is also referred to as the outside air temperature of the vehicle **20**. At least a portion of the temperature sensor **21** may be disposed outside the vehicle **20**.

(19) The electronic control unit **22** is an Electronic Control Unit (ECU of the vehicles **20**). The electronic control unit **22** controls various functions of the vehicle **20**.

(20) The in-vehicle device **30** is an information processing device. The in-vehicle device **30** is, for example, a navigation device. The in-vehicle device **30** includes a communication unit **31**, a positioning unit **32**, an output unit **33**, a storage unit **34**, and a control unit **35**.

(21) The communication unit **31** is configured to include at least one communication module that can be connected to the network **2**. The communication module is a communication module compliant with a mobile communication standard such as Long Term Evolution (LTE), 4th Generation (4G), or 5th Generation (5G), for example.

(22) The communication unit **31** may include at least one communication module capable of communicating with the temperature sensor **21** and the electronic control unit **22**. The communication module is, for example, a communication module corresponding to a standard of an in-vehicle network or a dedicated line.

(23) The positioning unit **32** can acquire position information of the vehicle **20** on which the in-vehicle device **30** is mounted. The positioning unit **32** is configured to include at least one receiving module corresponding to the satellite positioning system. The reception module is, for example, a reception module corresponding to Global Positioning System (GPS).

(24) The output unit **33** can output data. The output unit **33** includes at least one output interface capable of outputting data. The output interface is, for example, a display or a speaker. The display is, for example, a Liquid Crystal Display (LCD) or an organic Electro Luminescence (EL) display.

(25) The configuration of the storage unit **34** may be the same as or similar to the configuration of the storage unit **12**. The storage unit **34** stores data used for the operation of the in-vehicle device **30** and data obtained by the operation of the in-vehicle device **30**.

(26) The configuration of the control unit **35** may be the same as or similar to that of the control unit **13**. The control unit **35** executes processing related to the operation of the in-vehicle device **30** while controlling each unit of the in-vehicle device **30**.

(27) System Operation

(28) FIG. **2** is a sequence diagram illustrating an operation example of the system **1** illustrated in FIG. **1**.

(29) Processing of **S1** and **S2** may be performed by in-vehicle device **30** of any vehicle **20** of the plurality of vehicles **20** included in the system **1**.

(30) In **S1** process, in the in-vehicle device **30**, the control unit **35** acquires the data of the vehicle **20**. The data of the vehicle **20** may be any data as long as it is data related to the fuel efficiency of the vehicle **20**. In the present embodiment, the data of the vehicle **20** includes environmental data and traveling data of the vehicle **20**.

(31) The environment data of the vehicle **20** includes, for example, time-series data of the ambient temperature of the vehicle **20**. For example, the control unit **35** acquires time-series data of the ambient temperature of the vehicle **20** by sequentially receiving the data of the ambient temperature of the vehicle **20** detected by the temperature sensor **21** by the communication unit **31**.

(32) The travel data of the vehicle **20** includes, for example, time-series data of the position of the vehicle **20**, time-series data of the speed of the vehicle **20**, trip time data, and fuel consumption data for each trip time. In addition to or in place of these data, the travel data of the vehicle **20** may include historical usage data of the heating device, time series data of the temperature of the coolant of the engine, and time series data of the rotational speed of the engine. The temperature of the coolant of the engine is the so-called engine water temperature. The trip time is a time during which the vehicle **20** travels from the departure point to the destination. The fuel efficiency of the vehicle **20** is the travel distance of the vehicle **20** per unit capacity of the fuel. For example, the control unit **35** acquires time-series data of the position of the vehicle **20** by sequentially acquiring the position information of the vehicle **20** from the positioning unit **32**. The control unit **35** acquires time-series data of the speed of the vehicle **20**, data of the trip time, and data of the fuel consumption per trip time by communicating with the electronic control unit **22** via the communication unit **31**, for example. The control unit **35** acquires, for example, time-series data of the use history of the heating device, time-series data of the temperature of the coolant of the engine, and time-series data of the rotational speed of the engine by communicating with the electronic control unit **22** via the communication unit **31**.

(33) In **S2** process, the control unit **35** transmits the data of the vehicle **20** to the server device **10** via the network **2** by the communication unit **31**.

(34) In **S3** process, in the server device **10**, the control unit **13** receives data of the vehicle **20** from the vehicle **20** via the network **2** by the communication unit **11**. The control unit **13** stores the

received data of the vehicle **20** in the storage unit **12**.

(35) The process from **S1** to **S3** may be repeatedly executed for a predetermined time. The predetermined period may be set based on the traveling frequency of the plurality of vehicles **20** included in the system **1**. When the processes from **S1** to **S3** are repeatedly executed, the data of the plurality of vehicles **20** is accumulated in the storage unit **12** of the server device **10**.

(36) In **S4** process, in the server device **10**, the control unit **13** acquires the data of the vehicles **20** accumulated in the storage unit **12**. Based on the acquired data of the vehicle **20**, the control unit **13** specifies a setting condition for improving the fuel efficiency of the vehicle **20** by using the low-viscosity oil for each set area. An example of this processing will be described below.

(37) The control unit **13** calculates the fuel efficiency difference in the vehicle **20** by subtracting the fuel efficiency of the vehicle **20** when the high-viscosity oil is used from the fuel efficiency of the vehicle **20** when the low-viscosity oil is used for each set area. Here, the high viscosity oil may be an engine oil having an arbitrary viscosity as long as it has a viscosity higher than that of the low viscosity oil. The high viscosity oil may be selected depending on the low viscosity oil recommended for use. For example, it is assumed that OW-8 engine oil in the viscosity section of Society of Automotive engineers (SAE) is the low viscosity oil recommended for use. In this case, the control unit **13** may calculate the fuel efficiency difference in the vehicle **20** using OW-16 engine oil as the high viscosity oil, or may calculate the fuel efficiency difference in the vehicle **20** using OW-20 engine oil as the high viscosity oil. The control unit **13** may distinguish the vehicle **20** using the low-viscosity oil from the vehicle **20** using the high-viscosity oil based on the information on the type of the engine oil used for each of the plurality of vehicles **20** stored in the storage unit **12**. In addition, the control unit **13** may specify the vehicle **20** traveling in the set area based on the time-series data of the position of the vehicle **20** included in the traveling data of the vehicle **20**.

(38) The greater the difference in fuel efficiency in the vehicle **20**, the greater the degree of improvement in fuel efficiency of the vehicle **20** due to the use of low viscosity oil.

(39) Therefore, the control unit **13** specifies the environmental condition and the traveling condition of the vehicle **20** in which the difference in fuel consumption in the vehicle **20** increases, thereby specifying the setting condition in which the fuel consumption of the vehicle **20** is improved by using the low-viscosity oil. For example, the control unit **13** divides the environmental data and the traveling data of the vehicle **20** accumulated in the storage unit **12** into data groups. The control unit **13** calculates the fuel efficiency difference in each data group based on the information on the type of the engine oil stored in the storage unit **12**, thereby specifying the environmental condition and the traveling condition of the vehicle **20** in which the fuel efficiency difference in the vehicle **20** increases.

(40) For example, as illustrated in FIG. 3, the control unit **13** specifies a setting condition. In FIG. 3, the setting condition is specified for each setting region of “aaa” and “bbb”. In the setting area with “aaa”, the condition A is specified as a setting condition for increasing the fuel efficiency difference. In the condition A, the ambient temperature range from the temperature Ta1 to the temperature Ta2 is specified as the environmental condition of the vehicle **20**. In the condition A, a vehicle speed range from the speed Ra1 to the speed Ra2 is specified as the traveling condition of the vehicle **20**. In the setting area with “bbb”, the condition B is specified as a setting condition for increasing the fuel efficiency difference. In the condition B, the ambient temperature range from the temperature Tb1 to the temperature Tb2 is specified as the environmental condition of the vehicle **20**. In the condition B, the vehicle speed range from the speed Rb1 to the speed Rb2 is specified as the traveling condition of the vehicle **20**.

(41) In **S4** process, the control unit **13** may specify the setting condition for each vehicle type of the vehicle **20**. The structure and the like of the engine differ depending on the vehicle type of the vehicle **20**. Therefore, by specifying the setting condition for each vehicle type of the vehicle **20**, it is possible to accurately specify the setting condition for improving the fuel efficiency of the

vehicle **20** by use of the low-viscosity oil.

(42) After the processing of **S4**, the processing of **S5** and **S6** is executed. Processing of **S5** and **S6** may be performed by in-vehicle device **30** of any vehicle **20** of the plurality of vehicles **20** included in the system **1**. The in-vehicle device **30** performs **S5** and **S6** processes the same as or similar to **S1** and **S2** processes.

(43) In **S7** process, in the server device **10**, the control unit **13** receives data of the vehicle **20** from the plurality of vehicles **20** via the network **2** by the communication unit **11**.

(44) In **S8** process, the control unit **13** refers to the type of the engine oil in the storage unit **12**, and identifies the vehicle **20** that does not use the low-viscosity oil among the plurality of vehicles **20** that have received the data in **S7** process. Hereinafter, the vehicle **20** that does not use the specified low-viscosity oil is also referred to as “vehicle **20A**”. The control unit **13** executes a notification process when **20A** of vehicles received in the process of **S7** satisfies the setting condition. In the present embodiment, the control unit **13** executes the notification process when the environmental data and the traveling data of the vehicle **20A** satisfy the setting condition.

(45) For example, it is assumed that the control unit **13** specifies that the set area in which the vehicle **20A** travels is “aaa” as illustrated in FIG. **3**, based on the time-series data of the position of the vehicle **20** included in the travel data of the vehicle **20A**. Further, it is assumed that the vehicle speed included in the traveling data of the vehicle **20A** is in the range from the velocity **Ra1** to **Ra2**, and the outside air temperature included in the environmental data of the vehicle **20A** is in the range from the temperature **Ta1** to **Ta2**. In this case, the control unit **13** determines that the environmental data and the traveling data of the vehicle **20A** satisfy the condition **A** as illustrated in FIG. **3**.

(46) As an example of the notification process, the control unit **13** may transmit a notification recommending the use of the low-viscosity oil to the terminal device of the user of the vehicle **20A** via the network **2** by the communication unit **11**. The terminal device of the user of the vehicle **20A** may be a smart phone or an in-vehicle device **30** mounted on the vehicle **20A**. The terminal device of the user of the vehicle **20A** may cause the terminal device to display the received notification when receiving the notification recommending the use of the low-viscosity oil from the server device **10**. For example, when the terminal device of the user of the vehicle **20A** is the in-vehicle device **30**, the control unit **13** causes the display of the output unit **33** to display a notification recommending the use of the low-viscosity oil. By executing such a notification process, it is possible to recommend the use of low-viscosity oil to the user of the vehicle **20A**. Also, the user of the vehicle **20A** may not know the presence of low-viscosity oil. In this case, the user of the vehicle **20A** can be notified of the presence of the low-viscosity oil by executing the notification process. Here, the control unit **13** may transmit, to the terminal device of the user of the vehicle **20A**, the notification that recommends the use of the low viscosity oil and the information indicating the use effectiveness of the low viscosity oil. The information indicating the use effectiveness of the low viscosity oil is, for example, information indicating the degree of improvement in fuel efficiency of the vehicle **20A** due to the use of the low viscosity oil. In this case, the control unit **13** acquires the environmental data of the vehicle **20A** and the data similar to the traveling data received in **S7** processing from the traveling data and the environmental data accumulated in the storage unit **12** by the processing of **S3** from **S1**. The control unit **13** generates information indicating the use effect of the low-viscosity oil by calculating the fuel consumption difference in the vehicle **20** based on the obtained traveling data and environmental data. When the information indicating the use effectiveness of the low viscosity oil is transmitted to the terminal device of the user of the vehicle **20A** together with the notification, the use of the low viscosity oil can be strongly recommended to the user of the vehicle **20A**.

(47) As another example of the notification process, the control unit **13** may transmit the information of the user of the vehicle **20A** to the terminal device of the dealer of the engine oil by the communication unit **11**. With such a configuration, it is possible to recommend the use of low-

viscosity oil to the user of the vehicle **20A** via the vendor of the engine oil. The control unit **13** may transmit information indicating the use effect of the low-viscosity oil together with the notification to the terminal device of the dealer of the engine oil.

(48) Here, in **S8** process, the control unit **13** may not execute the notification process even when the environmental data and the travel data of the vehicle **20A** satisfy the setting condition, or when the travel data of the vehicle **20A** received in **S7** process satisfies the condition that causes the engine degradation. The condition causing the engine deterioration may be a condition that the number of times of travel of the trip time exceeding the predetermined time exceeds the predetermined number of times, or a condition that the traveling frequency at the vehicle speed exceeding the predetermined speed exceeds the predetermined frequency. The predetermined time, the predetermined number of times, the predetermined speed, and the predetermined frequency may be set in consideration of engine deterioration. In this way, the vehicle **20** can be protected by not executing the notification process when the traveling data of the vehicle **20A** satisfies the condition that causes the engine degradation.

(49) In this way, in the server device **10**, the control unit **13** specifies, for each set area, a setting condition under which the fuel efficiency of the vehicle **20** is improved by using the low-viscosity oil. The control unit **13** executes the notification process when the data of the vehicle **20** that does not use the low-viscosity oil satisfies the setting condition in the setting area. As described above, the degree of improvement in the fuel efficiency of the vehicle **20** due to the use of the low viscosity oil varies from region to region. In the present embodiment, since the setting conditions for improving the fuel efficiency of the vehicle **20** are specified for each setting area, it is possible to more accurately recommend the use of the low-viscosity oil to the user of the vehicle **20** who does not use the low-viscosity oil in the setting area.

(50) Although the present disclosure has been described above based on the drawings and the embodiments, it should be noted that those skilled in the art may make various modifications and alterations thereto based on the present disclosure. It should be noted, therefore, that these modifications and alterations are within the scope of the present disclosure. For example, the functions included in the configurations, steps, etc. can be rearranged so as not to be logically inconsistent, and a plurality of configurations, steps, etc. can be combined into one or divided.

(51) For example, in the above-described embodiment, the server device **10** as the information processing device specifies, for each set area, a setting condition for improving the fuel efficiency of the vehicle by using the low-viscosity oil based on the data of the plurality of vehicles **20**. In the above description, the server device **10** executes the notification process when the data of the vehicle **20** that does not use the low-viscosity oil satisfies the setting condition in the setting area. However, an arbitrary information processing device may specify such a setting condition and execute a notification process.

Claims

1. An information processing device comprising a control unit that specifies a setting condition for improving fuel efficiency of a vehicle by using low-viscosity oil based on data on a plurality of vehicles for each setting area set in advance, wherein the control unit is configured to execute a notification process to recommend a user of a vehicle in which the low-viscosity oil is not used to use the low-viscosity oil when data on vehicles in which the low-viscosity oil is not used meet the setting condition in the setting area.

2. The information processing device according to claim 1, wherein the control unit is configured to: calculate a fuel consumption difference by subtracting fuel consumption of the vehicle for a case where high-viscosity oil is used from fuel consumption of the vehicle for a case where the low-viscosity oil is used, and specify the setting condition by specifying an environment condition and a travel condition for the vehicle under which the fuel consumption difference is large; and

execute the notification process when environment data and travel data on the vehicle in which the low-viscosity oil is not used meet the setting condition.

3. The information processing device according to claim 1, wherein the control unit is configured not to execute the notification processing when travel data on the vehicle in which the low-viscosity oil is not used meet a condition under which engine deterioration is caused, even when environment data and travel data on the vehicle in which the low-viscosity oil is not used meet the setting condition.

4. The information processing device according to claim 1, further comprising a communication unit, wherein the control unit transmits a notification that recommends use of the low-viscosity oil to a terminal device of the user of the vehicle in which the low-viscosity oil is not used using the communication unit in the notification process.

5. The information processing device according to claim 1, further comprising a communication unit, wherein the control unit transmits information on the user of the vehicle in which the low-viscosity oil is not used to a terminal device of an engine oil dealer using the communication unit in the notification process.
